

Inductive Principle for R^* and R^+

Assume a set A , and any relations $R, Q \subseteq A \times A$.

$$\begin{aligned} & \forall a \in A. Q(a, a) \quad \wedge \\ & \forall a, a', a'' \in A. [R(a, a'') \wedge Q(a'', a') \implies Q(a, a')] \\ & \implies \\ & \forall a, a' \in A. [R^*(a, a') \implies Q(a, a')] \\ \\ & \forall a, a' \in A. [R(a, a') \implies Q(a, a')] \quad \wedge \\ & \forall a, a', a'' \in A. [R(a, a'') \wedge Q(a'', a') \implies Q(a, a')] \\ & \implies \\ & \forall a, a' \in A. [R^+(a, a') \implies Q(a, a')] \end{aligned}$$